

SonarQube Code Quality Report

Scribe - Notes App (MERN Stack)

10Pearls Shine Internship Program · Cohort 9 · Prepared by Saim Zafar

1. Overview

This report documents the static code analysis results produced by SonarQube (Community Edition, run locally via Docker) for both the backend and frontend components of the Notes App. Two separate projects were configured and analyzed: **notes-app-backend** (Node.js / Express) and **notes-app-frontend** (React / Vite). Both projects were scanned after implementation and again after addressing issues identified during automated review, including feedback from CodeRabbit.

2. Backend - notes-app-backend

The backend analysis covers the Express REST API, including authentication, notes CRUD logic, database access, middleware, and logging. The Quality Gate passed with no open issues across all measured categories.

Metric	Value
Quality Gate	Passed
Security	A (0 open issues)
Reliability	A (0 open issues)
Maintainability	A (0 open issues)
Duplications	0.0%
Security Hotspots	0

2.1 Overall Code Dashboard

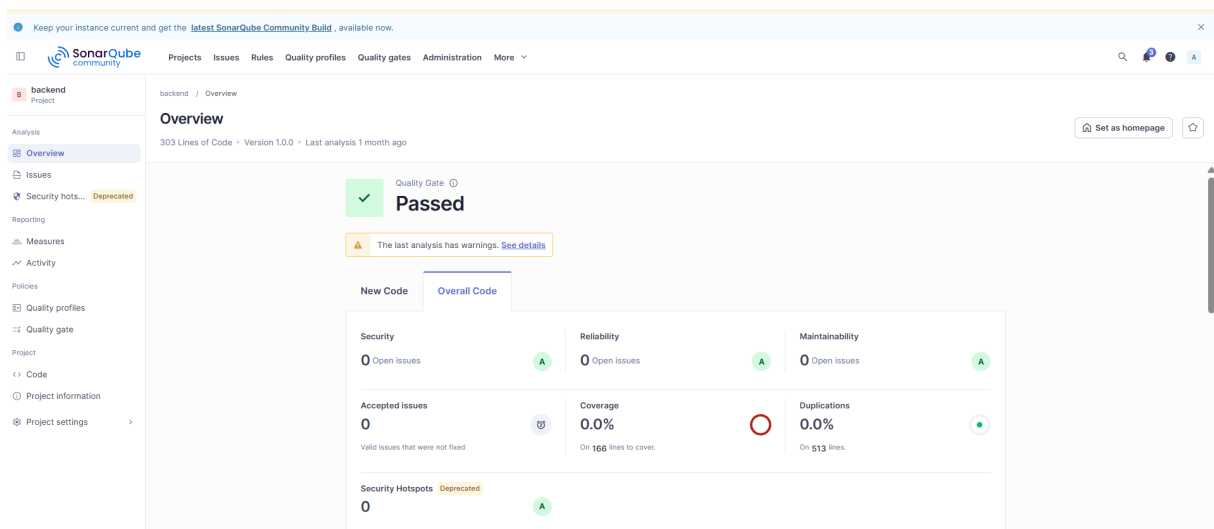


Figure 1. Backend project overview, Overall Code tab.

2.2 Issue Trend Over Time

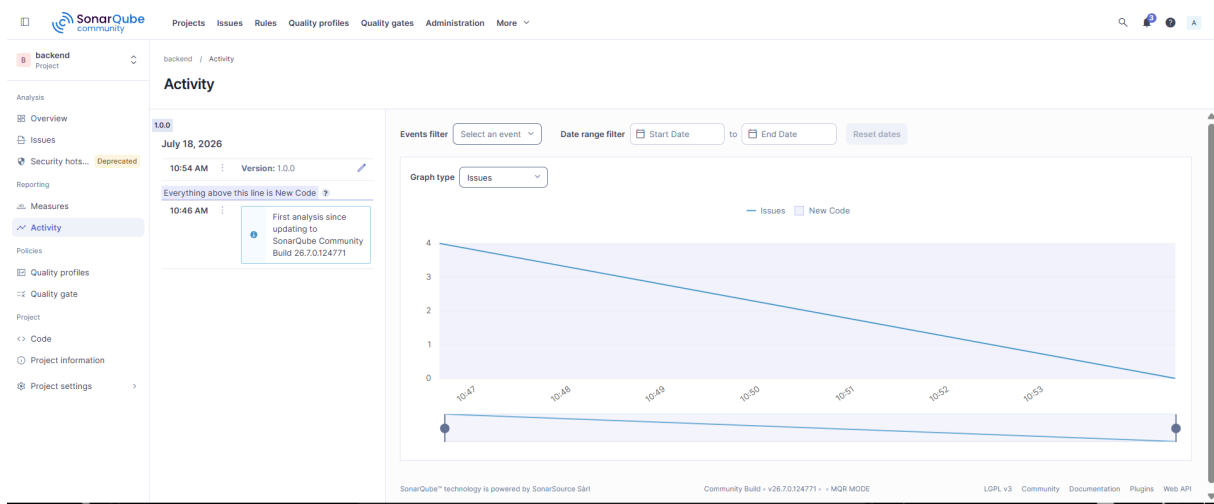


Figure 2. Backend issue count across successive scans, showing resolution of flagged issues over time.

3. Frontend - notes-app-frontend

The frontend analysis covers the React application, including authentication context, routing, the rich text editor, and page components. The Quality Gate passed with no open issues across all measured categories.

Metric	Value
Quality Gate	Passed
Security	A (0 open issues)
Reliability	A (0 open issues)
Maintainability	A (0 open issues)
Coverage	38.0% (on 200 lines)
Duplications	0.0%
Security Hotspots	0

3.1 Overall Code Dashboard

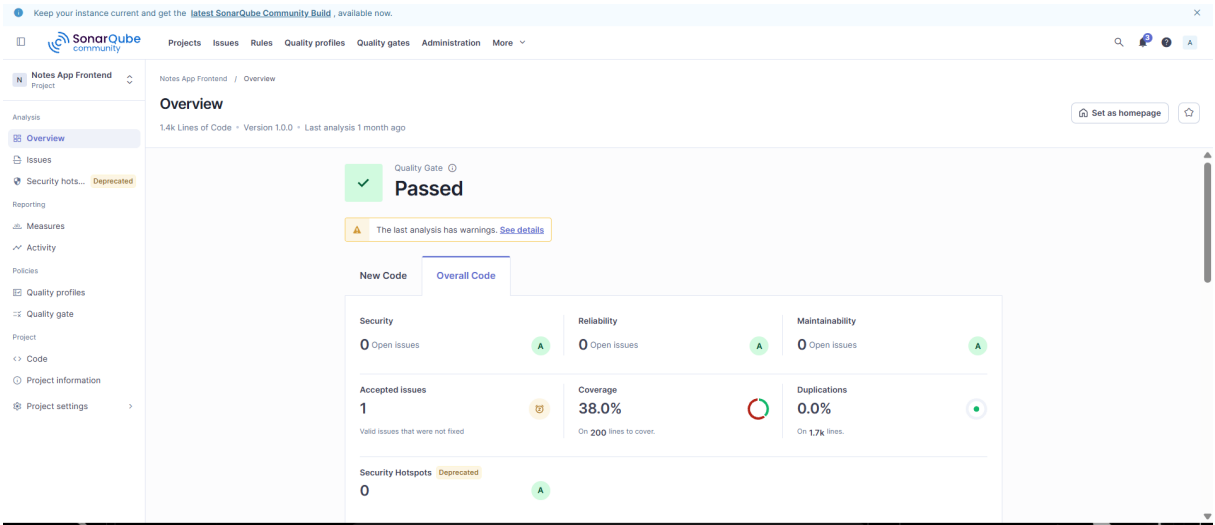


Figure 3. Frontend project overview, Overall Code tab.

3.2 Issue Trend Over Time

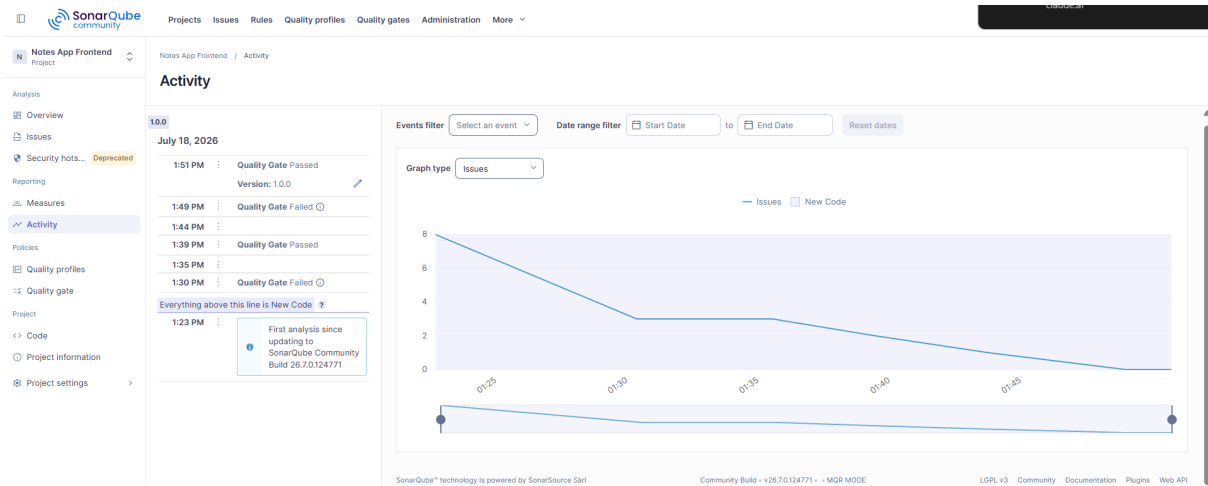


Figure 4. Frontend issue count across successive scans, showing resolution of flagged issues over time.

4. Summary

Both the backend and frontend components achieve a passing Quality Gate with an A rating in Security, Reliability, and Maintainability, and no code duplication. Issues identified during initial analysis and subsequent CodeRabbit review (including credential handling, input validation, error handling, and JWT verification) were addressed prior to this final scan.